CRICKET ANALYSIS

Submitted in partial fulfillment of the requirements
of the course **Innovative Product Development-III**

Year 3, Sem V Computer Engineering

By

Aditya Kathe	**60040230191**
Advaith Manoj	**60004230264**
Afroz Ansari	**60004230117**
Hasan Rampurawala	**60004230162**

Guide:

Dr. Kiran Bhowmick
Associate Head

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

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CERTIFICATE

This is to certify that the project entitled **"Cricket Analysis"** is a bonafide work of **"Aditya Kathe(60040230191), Advaith Manoj(60004230264), Afroz Ansari(60004230117) and Hasan Rampurawala(60004230162)"**submitted as a project work for the course **Innovative Product Development-III, Year 3, Semester V, TY B.Tech Computer Engineering**

Dr. Kiran Bhowmick
Internal Guide

Dr. Meera Narvekar
Head of Department

Dr. Hari Vasudevan
Principal

IPD Project Report Approval for BTech Semester V

This is to certify that the project entitled **"Cricket Analysis"** is a bonafide work of **"Aditya Kathe(60040230191), Advaith Manoj(60004230264), Afroz Ansari(60004230117) and Hasan Rampurawala(60004230162)"** submitted as a project work for the course **Innovative Product Development-III, Year 3, Semester V, TY B.Tech Computer Engineering**

Examiners

1.----------

2.----------

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Aditya Kathe 60040230191

Advaith Manoj 60004230264

Afroz Ansari 60004230117

Hasan Rampurawala 60004230162

Date:

Abstract

The integration of advanced software tools and data analytics has transformed team strategies, player performance evaluation, and match outcome prediction in sports like cricket. This report examines real-world applications from football—such as Leicester City's Prozone-driven tactical optimization for their 2015-16 Premier League win and Liverpool's use of StatsBomb for high-pressing strategies—and adapts them for T20 cricket analytics.

Existing cricket analytics often overlooks recent player form, venue-specific condition ,relying on raw stats without contextual impacts. These gaps lead to subjective team selections and suboptimal in-game decisions, as noted in the project's problem formulation.

The project introduces Win Probability Added (WPA) and Runs Probability Added (RPA) models to quantify each balls contribution to victory and scoring, aggregated into z-scored player metric (bat_rpa_z, bowl_wpa_z) for fair comparisons across roles. These feed into a 4D player vector [bat_rpa_z, bowl_rpa_z, bat_wpa_z, bowl_wpa_z], enabling cosine similarity for replacements and confidence-weighted career values blending 80% long-term data with 20% rolling form.

End applications include Player vs. Player (PvP) for matchup edges, Player vs. Team (PvT) for opposition analysis, synergies scoring for batting/bowling pair (e.g., RPAtogether - solo baselines), and Best XI optimizations using editable constraint like min/max batter. Implemented via FastAPI backend and Flutter WebApp, these tools support decisions like optimal opening pairs or bowler rotations.

Built on Python with ball-by-ball data from Cricsheet, the system processes raw inputs into structured metrics (RMSE 25.12 for RPA, ROC-AUC 0.8679 for WPA), stored for API access without real-time video or injury tracking. This data-driven approach bridges traditional cricket intuition with explainable ML, enhancing win probabilities for IPL-style scenarios.

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List of Abbreviations

Sr. No.	Abbreviation	Expanded form
i	T20	Twenty20
ii	ODI	One Day Internationals
iii	ICC	International Cricket Council
iv	WTC	World Test Championship
v	WPA	Win Probability Added/Analysis
vi	RPA	Runs Probability Added / Runs above Expected
vii	PvP	Player vs Player
viii	PvT/PvT	Player vs Team
ix	GUI	Graphical User Interface
x	API	Application Programming Interface
xi	RMSE	Root Mean Square Error
xii	MAE	Mean Absolute Error
xiii	R^2	R-squared
xiv	ROC-AUC	Receiver Operating Characteristic - Area Under Curve
xv	xG	Expected Goals
xvi	IPL	Indian Premier League
xvii	BBL	Big Bash League



12TH MAN

1. Introduction

Cricket is the most popular sport in south Asia, which is why in this report we discuss methods of analysing this sport and the performance of players. There are three main formats of cricket that are played internationally: Test cricket, Twenty20 (T20) cricket and One Day Internationals (ODI) and there are international tournaments organized by the ICC for these formats only, which are the World test championship. ODI world cup, champion's trophy and the T20 world cup.

In a game of cricket, an inning is the duration of play in which one team bats a set amount of overs and the opposing team bowls the set amount of overs. An over is made of six balls, after which the bowler changes the side of the pitch they bowl from. A test match consists of four innings and both ODI and T20 consist of two innings. The goal of the team that bats second is to score more runs than the team that batted first and the goal of the team that batted first is to defend its score in the second innings.

Each team has eleven players and each team must have a wicket keeper. Players can be broadly classified as three types: batsman, bowler and all rounder who can both bat and bowl. The wicket keeper also doubles as a batsman. The performance of a batsman is measured by the number of runs scored, the average number of runs scored throughout their career, and the strike rate at which they score runs (ratio of runs scored and balls faced). The performance of a bowler is measured by the number of wickets taken, the number of runs conceded and economy of the bowler (ratio of runs conceded to the number of overs bowled). The performance of an all rounder is measured by both their batting and bowling.

The primary goal of this project is to develop a system that uses statistical analysis and machine learning to recommend a team of players that are most suited to play against a particular team at a game of cricket. The aim is to use historical statistics of players' performance against other teams and players, data specific to venues where the match is played like, average number of runs scored at a particular venue, average number of wickets taken etc. to recommend a team of players and to predict the outcome of the match.

Most research done in this area uses data which do not reflect the recent developments in strategies which teams use and also do not reflect the change in rules of the game. Most research also uses a small training and test set as well like matches played in a particular tournament. This project uses a large number of matches not limited to a particular tournament.

This product emphasises on use of extensive and broad spectrum data that considers a wide range of attributes to cover all aspects of a player's performance metrics and its impact on the team. It also considers the team's performance and how it varies due to a specific player. This approach helps us in finding the BestXI with the highest probability of winning a match. It also enables the replacement of a specific player to find a more suitable and eligible candidate(team member) that would enhance the team's performance even further.

This approach is better than the current techniques that solely rely on regression, progression and predictive analytical methods as there is no promise of a fixed objective result of victory and the model enhances to give better results than the previous iteration or combination.

2.NEED OF THE PRODUCT

2.1 Why is the Product Needed?

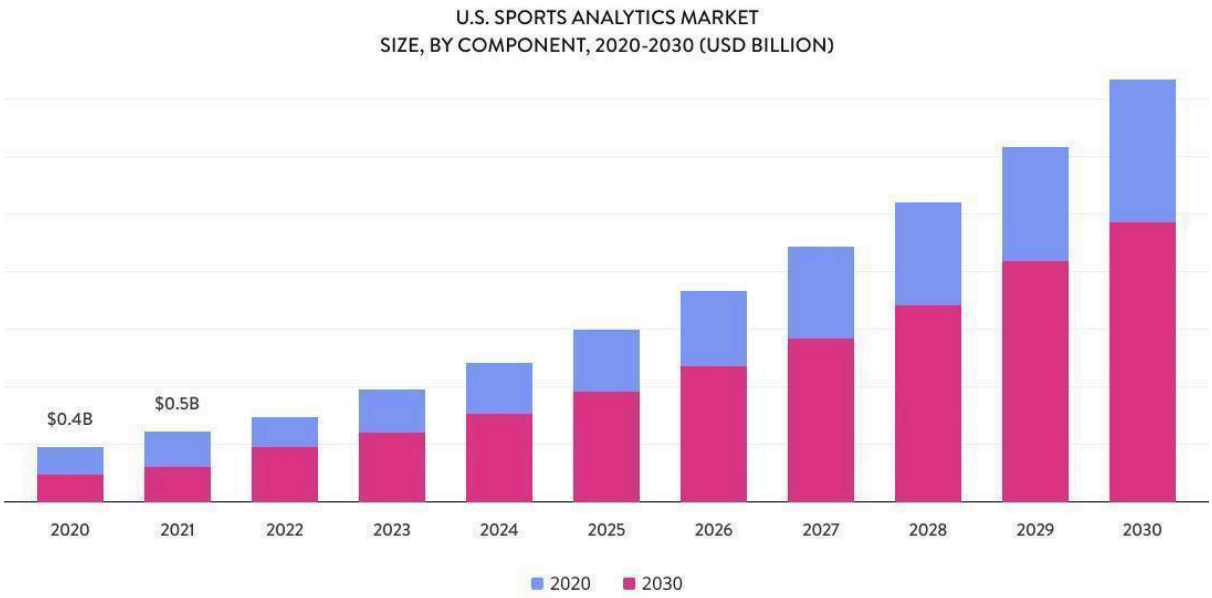


Fig. 2.1. Market growth of Sports Analytics in USA

The sports analytics industry is expanding rapidly. Referring to Figure 2.1, this shows that sports analytics is becoming increasingly dependent on data-informed models. The sports analytics industry in the U.S. is also expected to grow substantially during the period from 2020 to 2030. This shows that sports analytics is shifting away from models that are dependent on human intuition to models that are based on objective sports analytics. The sports analytics industry is also expected to provide support in processing vast amounts of match information to provide intelligence.

Despite the fact that there is such growth, there still exists a gap in domain-specific, explainable analytics tools for the sport of cricket, specifically those that go beyond the level of information presented by the plays on the surface. Current systems tend to have a lack of understanding of context related to the plays. There is a resulting requirement for such a system to incorporate all the analytics into one place.

The proposed product fills this requirement by utilizing advanced statistical modeling and data pipelines to generate insights such as Player vs. Player comparison insights, Player vs. Team comparison insights, Win Probability Analysis insights, as well as insights for optimal Team Selection.

2.2 Analysis of Existing Software

1. Wyscout

Application: Scouting and match analysis platform.
Features:

- Extensive player database and match footage.
- Heatmaps, pass maps, and statistical breakdowns.
- Tactical and technical breakdowns.
- Used by scouts and coaches for recruitment and planning.

Disadvantages:

- Mainly for scouting rather than deep tactical analysis.
- Subscription cost can be high for smaller organizations.
- Limited real-time functionality.

2. Stats Perform (Opta)

Application: Advanced statistical and AI-driven insights.
Features:

- Real-time and post-match data.
- xG, possession chains, and detailed player tracking.
- AI-based predictive analysis.

Disadvantages:

- Expensive and enterprise-focused.
- Data can overwhelm smaller coaching staffs.
- Requires technical expertise to interpret effectively.

2.3 Applications of the Product

12th Man is AI/ML based **Cricket Analytics and Decision-Support System** that helps professional experts like coaches and analysts assist teams, analyse train and strategise in evaluating player performance, team composition, and match outcomes using data-driven models. The project uses historical and modeled performance information into clear, explainable recommendations at both player and team levels.

The product mainly provides the assistance in the following aspects of Cricket Analytics:

- **Team Combination Evaluation**
- **Player v/s Player**
- **Players v/s Team**
- **Team performance optimization**
- **Player selection and replacement analysis**
- **Match situation awareness through win probability metrics**

The project uses micro-level tools like Players v/s Player and Player v/s Team to analyse and provide insight on interactions like how a batsman fares against a particular bowler or against the entire team. PvP analysis is applied whenever staff need to understand individual duels that could decide key moments in a match. By studying historical and modeled contests between batter–bowler pairs, the system helps identify favorable and unfavorable matchups, potential weak links to target, and bowlers or batters who are statistically well-suited to particular opponents. Similarly a PvT helps in enhancing the decision making for a selector to decide how a player would perform in an upcoming match. This makes PvP and PvT a building block for planning over-by-over strategies, field settings, and bowling changes that exploit specific micro-advantages.

Use of macro-level tools like Win Probability Analysis (WPA) is applied to primarily track how likely each side is to win at different points of the match based on the current state. It determines how far a team is away from winning a match helping analysts understand and identify the key momentum shifts from a major moment in the match like taking key wickets or achieving major partnerships. Analysts can judge on how much improvement can be made.

The project also helps in finding the synergy analysis which evaluates how combinations of players perform together rather than individually. This helps us in deciding partnerships of players in an inning and opening lineups to increase the WPA output. This feature captures non-linear team dynamics that individual metrics cannot explain therefore helping in solving complex synergy metrics.

Another feature of the project is to find a player replacement and perform similarity analysis. This feature identifies suitable replacements for a player based on similarity metrics and role compatibility thus helping WPA in finding out a better combination of players that would increase the performance of the team, maintaining squad resilience over long season and multi-tour formats. Particularly helpful when a player is unavailable or out of form, the system identifies a set of players with similar skills that contribute towards the team in the player's stead.

The Best XI Recommendation feature operationalizes all these analytical outputs into concrete team selection options. It suggests optimal playing XIs based on opponent conditions, player roles, synergies, and the WPA associated with different team combinations generated from available squads. This is applied directly in selection meetings to narrow down choices, justify selections with data-backed explanations, and align the final XI with the highest projected win probability rather than purely subjective judgment.

Across all these features, the product is applied to deliver objective performance evaluation, reduce human bias, and support transparent, explainable decision-making. Teams use its extensive time-series analysis to track players over formats, compare options for different match types, and refine long-term development plans for individuals. As a result, the system is suitable not only for domestic and international teams but also for cricket academies, analysts, coaching staff, and research-oriented sports analytics projects that seek to improve individual performance, optimize team composition, and design data-driven match strategies in a consistent, repeatable manner.

3.Survey

3.1 Case Study Survey

Here are some real-world examples of how using these tools impacted team and player performance:

1. Leicester City – Premier League Champions (2015-16)

Tool Used: Prozone (performance analysis)

- **Impact**:
    - Leicester City used Prozone for detailed match analysis to optimize their tactics and player usage. By focusing on efficient pressing, counter-attacking strategies, and maximizing player strengths, they defied the odds to win the Premier League title.
    - Players like Jamie Vardy and Riyad Mahrez benefited from insights that highlighted their roles in transitions and attacking phases
    - The team maintained top physical performance, utilizing data to prevent injuries and ensure peak fitness.

2. Liverpool FC – Champions League and Premier League Success (2019-20)

Tools Used:

- StatsBomb (data analytics)
- Catapult Sports (GPS tracking and fitness management)

- **Impact**:
    - Jurgen Klopp's staff integrated data from StatsBomb and Catapult to develop high-pressing strategies and track player workloads.
    - Players like Trent Alexander-Arnold and Andrew Robertson optimized their positioning and crossing accuracy, contributing to Liverpool's attacking and defensive dominance.
    - Result: Liverpool won the Champions League in 2019 and the Premier League in 2020.

3. Brentford FC – "Moneyball" Approach

Tools Used: Smartodds and SciSports (analytics and recruitment)

- **Impact**:
    - Brentford embraced a data-driven recruitment strategy, identifying undervalued players who fit their tactical system.
    - Example: Signing Ollie Watkins and Ivan Toney, who excelled after joining.
    - Brentford rose from the Championship to secure a stable position in the Premier League, outperforming clubs with larger budgets.

4. Spain National Team – Euro 2012

Tool Used: Mediacoach (match analysis)

- **Impact**:
    - Spain used Mediacoach to analyze their ball possession strategy and ensure players maintained compact formations.

- Insights from Mediacoach allowed them to fine-tune their tiki-taka style, leading to dominance in Euro 2012.

5. FC Midtjylland – Danish Champions

Tools Used: StatsBomb and bespoke in-house analytics

- **Impact**:
  - The club employed data analytics to develop a niche strategy of maximizing set-piece opportunities, resulting in an unusually high percentage of goals scored from dead-ball situations.
  - Midtjylland consistently performed above expectations in domestic and European competitions.

3.2 Literature Survey

Elo rating systems [1]

A system to measure the current prowess or strength of an individual or a team is established, represented by a numeric value . A rating of 1500 is chosen as the initial rating for players which comes from the original implementation of Elo in chess, where an Elo rating of 1500 represents an average player.

The most important use of Elo is in outcome prediction either between two players or two teams. The method used in the framework of Elo is the same used here i.e.,

$$E(A) = \frac{1}{1 + 10^{(R_b - R_a)/400}}$$

and the rule to update Elo rating is,

$$R_{new} = R_{old} + K(S - E)$$

Glickman establishes a system which enables the user to control change in Elo as uncertainty in outcome of a match decreases [2]. This parameter estimation is used to quickly estimate the Elo of new players or teams and to decrease the change of Elo of older or more established teams. The value of the parameter K is calculated using the formula [3],

$$k = \frac{q}{\sqrt{1 + M/c}}$$

where q is a base value to represent volatility, M is the number of matches played and c is a scaling constant.

Further,

Runs per wicket [4], [5]

Lemmer argues for runs per wicket (RPW) as a more effective measure of batting performance [3][9]. It gives value to the amount of wickets lost, and runs scored in a set amount of wickets. The loss of a wicket can be a turning point and a major deciding factor in the outcome of a match. It results in re-establishing the partnership and a new player will require time to set. Lemmer argues that RPW represents the batting performance of a team more accurately than measures like total runs, batting average, strike rate etc.

Performance indicators [6]

This paper tries to establish factors or measures of batting, bowling and fielding performance which most significantly affect the outcome of a cricket match [5]. Performance indicators like total number of balls faced, total number of runs scored, total number of wickets taken, etc. Is used. The significance of a performance indicator is measured by the ratio of the difference between the two means of that performance indicator for the two teams and the standard deviation.

Match outcome prediction [7]

A system to recommend a team and predict the outcome of a game of cricket is discussed in this paper. A ranking of players is created based on performance measures relative to other players playing in a tournament like the world cup. On the basis of these rankings, players are clustered using K Nearest Neighbours and a system to recommend playing XI is made. Outcome of the game is predicted using support vector machines.

Outcome of the survey

A system to measure the performance of players against specific players and teams can be made using Elo rating of players and teams, runs per wickets, win probability and win probability analysis to recommend players against a team.

Machine learning techniques can be used to predict the outcome of a cricket match by considering player, team and venue statistics.

4.Problem Formulation

4.1 Problem Formulation

The following are the issues that arise when teams don’t use analytical data:

When analytical data sets and techniques are not utilized in cricket team planning, several problems can arise, impacting both team and individual performance. Here's an overview:

1. Lack of Objective Decision-Making

- Selection Bias: Team selection may rely heavily on subjective opinions or intuition rather than performance trends.

2. Inadequate Match Preparation

- Poor Opposition Analysis: Without data, teams may fail to identify the strengths, weaknesses, and patterns of opposing players and teams.

3. Limited Player Development

- Undefined Progress Metrics: Without data, tracking a player’s progress becomes difficult, which may lead to inadequate coaching or misidentification of areas needing improvement.
- Missed Potential: Emerging talent may not be identified if performance is judged solely on qualitative assessments.

4. Ineffective In-Game Adjustments

- Delayed Reactions: Teams may struggle to adapt strategies in real-time without insights into evolving match conditions and player performance.
- Missed Opportunities: Incomplete data makes it harder to spot trends or predict outcomes, such as which bowler to deploy at a crucial moment.

6. Inconsistent Performance Monitoring

- Subjective Evaluations: Player form and assessments may rely on subjective judgment rather than measurable indicators.
- No Benchmarking: Teams might lack benchmarks to compare their performance against competitors or historical data.

In this project, we rely solely on data and information and patterns derived from which provides objective information. It provides data about player statistics before every match and gives extremely accurate and relevant data about players in points of time for analysis. Ball-by-ball data of matches allows us to predict and analyse how the outcome and strategies of teams should change based on current conditions during a match.

4.2 Product Objectives

Core Features: Decide on the primary functionalities this software will offer. Common features include:

- Performance analysis (player movements, strengths, weaknesses)
- Data visualization dashboards

Objectives of this product :

The use of analytics and software tools in cricket can closely mirror their application in football, with modifications tailored to cricket's unique requirements. By leveraging similar technologies, cricket teams can improve player performance, strategic decision-making, and injury prevention. Here's how this can be projected:

1. Performance Analysis

Cricket teams can use tools to analyze player and team performance during matches and training sessions.

- **Key Metrics**:
    - Batting: Strike rate, boundary percentage, dot-ball percentage, and performance under pressure.
    - Bowling: Line and length accuracy, economy rate, wicket-taking ability in various conditions.
    - Venue , team strength(elo) and historical climate
- **Example in Action**:
  The Indian Premier League (IPL) teams, such as the Mumbai Indians, use detailed match and player analytics to strategize against opponents and optimize player roles.

2. Player Selection and Recruitment

Data-driven scouting can identify talents suited for specific formats or conditions.

- **Key Approaches**:
    - Analyze domestic cricket statistics to identify players who excel under high-pressure

3. Tactical Planning and Opponent Analysis

Cricket teams can use tactical software to plan strategies for specific opponents.

- **Key Metrics**:
    - Opposition's strong and weak areas (e.g., bowlers struggling with particular deliveries or batters vulnerable to spin).

4. Team Selection Optimisation and Match Outcome Prediction

Advanced analytics can support coaches and analysts in selecting the best possible playing eleven.

Key Approaches:

Evaluate how the presence or absence of specific players affects team stability, match pressure handling, and overall match outcomes.
Use machine learning models to simulate match scenarios with different player combinations.
Incorporate past performance, recent form, pitch behavior, weather conditions, and opposition strengths to predict match outcomes.
Assess chemistry between players such as opening partnerships or bowler combinations that consistently perform well under certain conditions.

Tools:

Predictive modelling tools using machine learning algorithms such as Random Forest, XGBoost, and neural networks.
Performance dashboards that visualize the expected contribution of each player to the match result.

Example in Action:
Teams in tournaments like the IPL and BBL increasingly rely on data-driven match simulations to select balanced squads and forecast the probability of winning with different lineups.

4.3 Novelty

The novelty of this system is that it combines various statistical aspects and makes them work together to get advanced players and match analytics . The WPA and RPA applied and made to work together gives us the actual player performance relative to the outcome of the match , instead of just basing quality of a player on raw statistical data .

Temporal elo rankings of the teams are also calculated and applied in the WPA -RPA model , which lets us reward players making positive contributions in a weaker team against a strong team and also punish the players in strong teams underperforming against weak teams .

This model also lets us filter out the performances of batters and bowlers after the match is mathematically lost (i.e. in chases when req rr gets more than 36) . This applications use balanced weights of both historical and rolling form data so that the brilliance of the player and current form of the play are both considered . Confidence of the date i.e. the volume of a particular data in the database is also considered in final valuations , so players with richer experience are rewarded .

Because WPA and RPA are additive in nature, players of specific domestic competition or nationality can benefit or lose due to abundance or lack of data , for that we cap the data for these metrics in taking historical data in applications. We have also taken historical venue data , average first and second innings scores , phase wise scores, runs wickets .

Broad sport decisions like xi selection against team , synergy match , player replacement can be made with the help of the model as well as in match decisions like what batsman should be sent at a particular stage of a match given the venue , bowling lineup as well as who bowls a particular over can be made.

For BestXI we have used editable constraints like min/max number of batters or bowlers and all rounders.Other than that we also can list out 3 or 4 extra players to also serve the impact player rule.All these players are selected on the virtue of their historical performance and contributions as well as their current T20 form along with their sum performance against their entire squad. Venue can also be optionally stated to improve precision.

The historical WPA - RPA computation data enables to further expand into more applications in future.

4.4 Scope of the Project

This project is the development of a data-driven analytical system that is based on match wise data of Cricket.

Coverage:

- The system analyzes the players performance for both batting and bowling using the structured matches and ball by ball dataset.
- The system analyzes the teams performance while taking into consideration all the players in that team.
- The data in use for the project has been restricted to the professional T20 format of the game, which includes international fixtures as well as domestic leagues (IPL).
- The model takes into consideration venue, player's recent form and opposition strength to provide strong analytics on field like player versus player, player vs team, team vs team, and also suggest the best playing XI against a given team.

Limitations:

- The model does not provide real-time match analysis and relies on historical match data to provide an outcome.
- Video-based analysis like player movement tracking are outside the scope of the project.
- Determination of fielding positions (like wicket keeping, slips, mid-wicket) are not taken into consideration for the project.
- There is no consideration for injury of a player due to lack of medical diagnostic data.
- The importance of matches like playoffs and finals in a league and rivals in international fixtures are not in consideration of the model.

Assumptions:

- All players are assumed to be fit to play and be ready for selection.
- Sudden changes in climatic factors during a match are not updated.

The scope ensures that the project is technically feasible within the given academic timeframe while providing insightful analytics to help boost a team's performance.

5.Proposed Design

5.1 Proposed Model

The proposed design can be divided into four phases:

Data sorting:
Extract data from the web which is in the form of ball by ball delivers or match scorecards. Create a list of scorecards for every match played in a particular stadium.

Player Quantification and feature extraction

- Generate metrics to rank players for particular venues. Example of metrics that can be used:
- Batting Average
- Batting Strike Rate
- Milestone Reaching Ability
- Outrate
- Boundary runs per innings
- Then, normalize them using general metrics for that venue like:
- General Batting Average
- General Batting Strike Rate
- General Outrate

Player ranking and clustering:
Rank the players according to player qualifications, and cluster them.

Playing XI recommendation and predicting the outcome of a match:
Use K-Nearest Neighbors to recommend a playing XI on based on the recommended players, compare the players of both the teams to predict the outcome of the match.

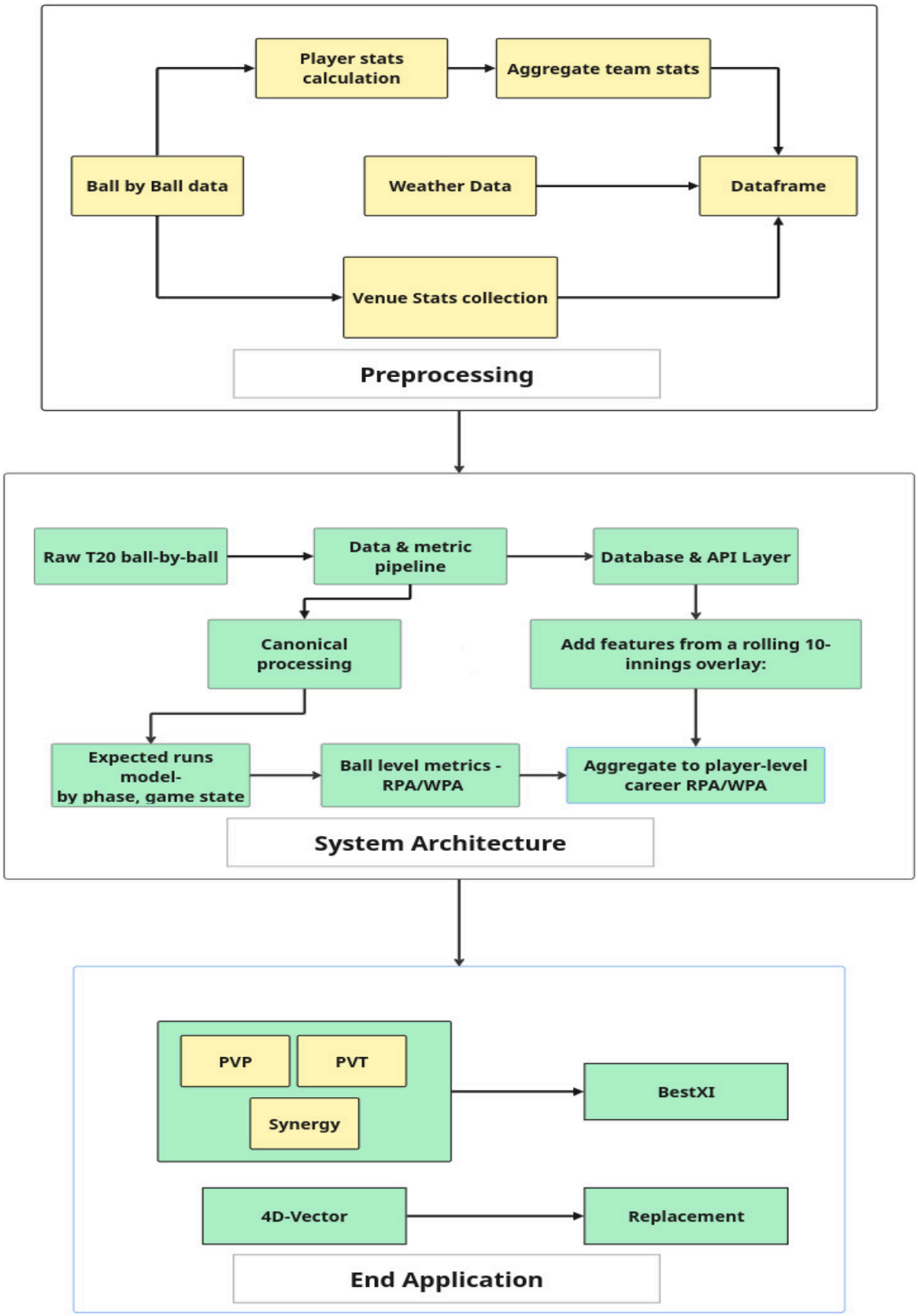


Fig. 5.1. Proposed Design Workflow

1. **Preprocessing Layer:**

   In this layer the raw data is fetched from the CricSheet[8] website via a webscraper and raw inputs are converted into structured valid inputs.

   Ball-by-ball data of every delivery bowled in the match is recorded to prepare for further stats calculation.Using this data **individual stats** for all players participating in the match is calculated. The aggregate of all the individual stats is then used to calculate **Aggregate Team stats** of the entire support team. This output is then added to the Dataframe for further procedures. For better understanding of the match we also gather the data related to the venue of the match like the pitch and weather conditions and their situations. **Venue Stats collection** is gathered and put into the dataframe. Weather in the region of the venue at the time of the match is also fed into Dataframe.

   This helps us in gathering all the attributes that affect the performance of an individual player and their team's performance in the entire match.

2. **System Architecture**:

   In this layer we design the system such that the product can perform real analytical computations and transform the data into a format that can derive performative and meaningful metrics.

   Raw T20 ball-by-ball data refers to the standardised data that will be used to derive meaningful metrics. The **Data & metric pipeline** converts the raw data that enter the system into data format suitable to compute certain analytical computations. **Canonical processing** cleanses and standardizes by resolving naming conventions, resolving inconsistencies, and deriving match context such as innings, overs remaining, wickets in hand, and match phases. **Expected Runs Model (by Phase and Game State)** estimates the expected number of runs for each delivery based on match conditions like phase of play, overs remaining, and wickets available. This provides context for further baseline operations. **Ball-Level Metrics (RPA and WPA)** converts each delivery into impact metrics such as Runs Probability Added (RPA) and Win Probability Added (WPA), quantifying how much a ball contributes to scoring potential and chances of victory. **Aggregate to Player-Level Career RPA/WPA–**Ball-level impact metrics are aggregated to generate career and innings-level performance summaries for players, enabling long-term and role-based evaluation in a team. **Rolling 10-Innings Feature Overlay** incorporates rolling windows that adjust metrics based on recent performances, confidence levels, exposure, and batting positions, ensuring that evaluations reflect current form. All processed metrics are stored in databases and are fetched via APIs, allowing efficient access by the frontend application without recomputation.

3. **End Application Layer**:

   Player vs Player (PvP) analyzes head-to-head matchups between players, such as batter versus bowler effectiveness, using aggregated impact metrics. **Player vs Team (PvT)** evaluates how a player performs against an entire opposing team, helping identify opponent-specific strengths or weaknesses. **Synergy Analysis** measures how combinations of players perform together, capturing partnership strength and complementary skill interactions. These attributes together give us the BestXI i.e., the best possible combination of 11 players that would give the best performance possible and the highest chance of victory. **4D Player Vector** represents each player using a multi-dimensional vector capturing batting, bowling, form, and contextual impact, enabling similarity and replacement analysis. This is used to identify suitable replacements for players already in the team to find more suitable or eligible candidates.

6. IMPLEMENTATION

6.1 GUI Design

The project is developed as a backend-oriented analytics system using FastAPI. It provides RESTful APIs; the documentation and visualization are provided by Swagger UI, considering the main objective of the analytical services provision.

Swagger UI is an interactive graphical interface that allows users to:

- access the various analytics endpoints available
- retrieve answers in a JSON output format enabling analysts, developers, and researchers to interact with this system without the need to develop a frontend application.

Images included below are illustrations of Swagger-based graphical user interface along with endpoint descriptions, request schemas, and sample outputs.

The implemented list of API endpoint include:

- Player Replacement : provides a list of best possible players to replace the current input player.
- Player vs Player : judges and compares the analytics of a bowler and a batsman with additional conditions like venue to provide an analysis of which player will come out on top.
- Player vs Team : judges and compares the analytics of an individual player against a team to provide insight on how well their performance is against said team.
- Player Synergy : provides insight on how well two batsmen or two bowlers synergize with each other with a minimum level of confidence.
- Best XI Optimizer : provides the best team of 11 players given input on players of the opponent team and available players of the team to decide from.
- Metadata: provide all the analytical details of players and teams along with the player's elo score.

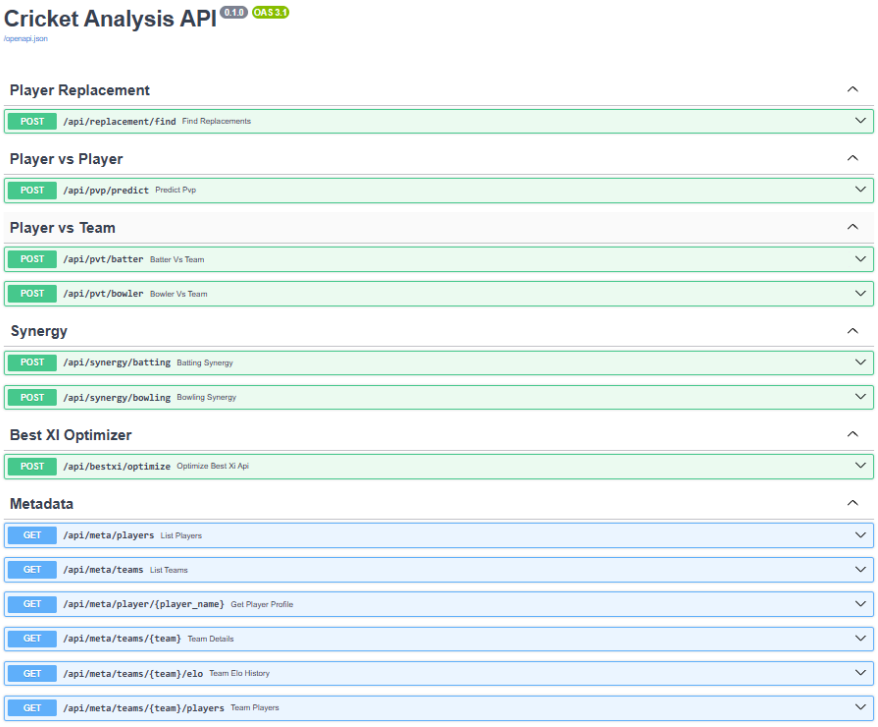


Fig. 6.1. List of API Endpoints via Swagger UI

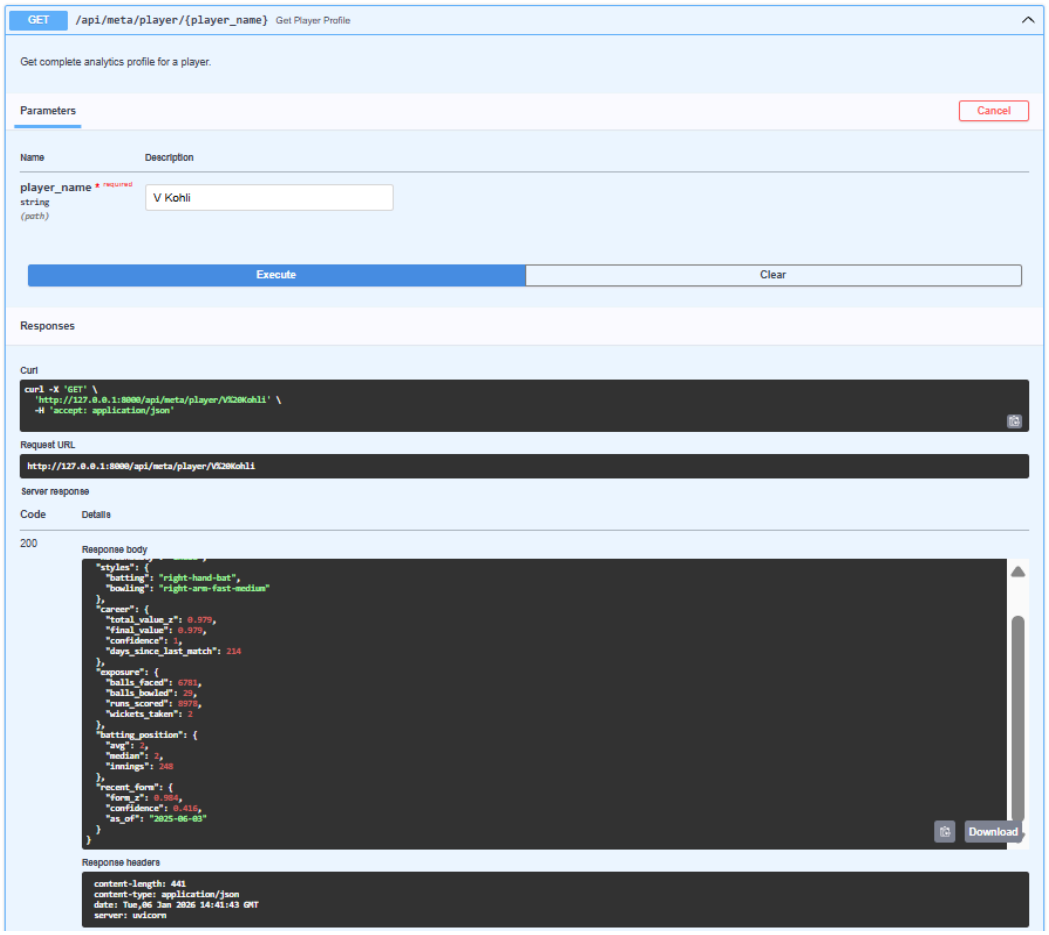


Fig. 6.2. Analytics of player on request of player name (V Kohli)

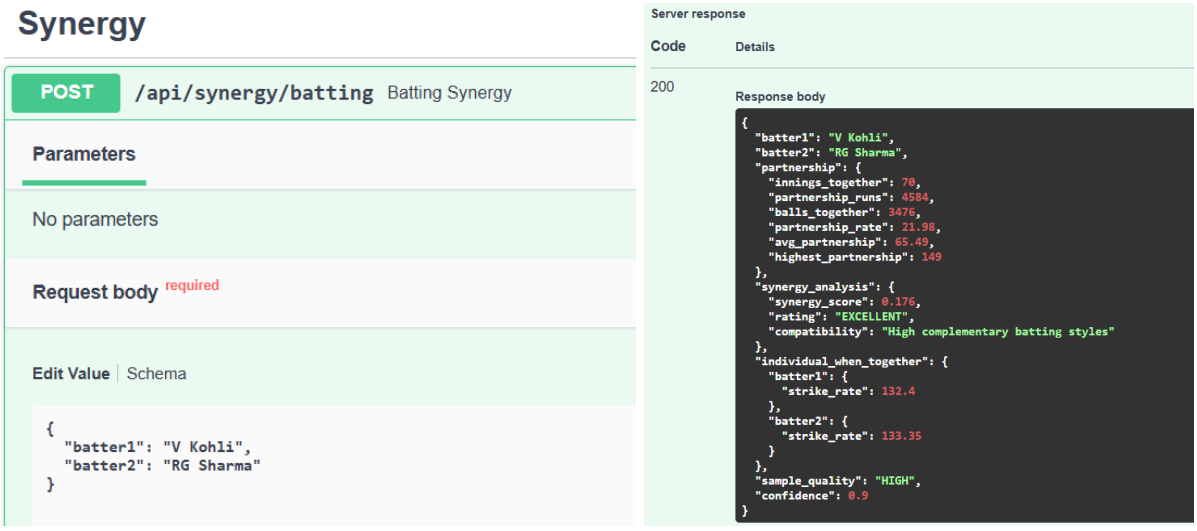


Fig. 6.3. Synergy of two batsmen on request of player names (V Kohli and RG Sharma)

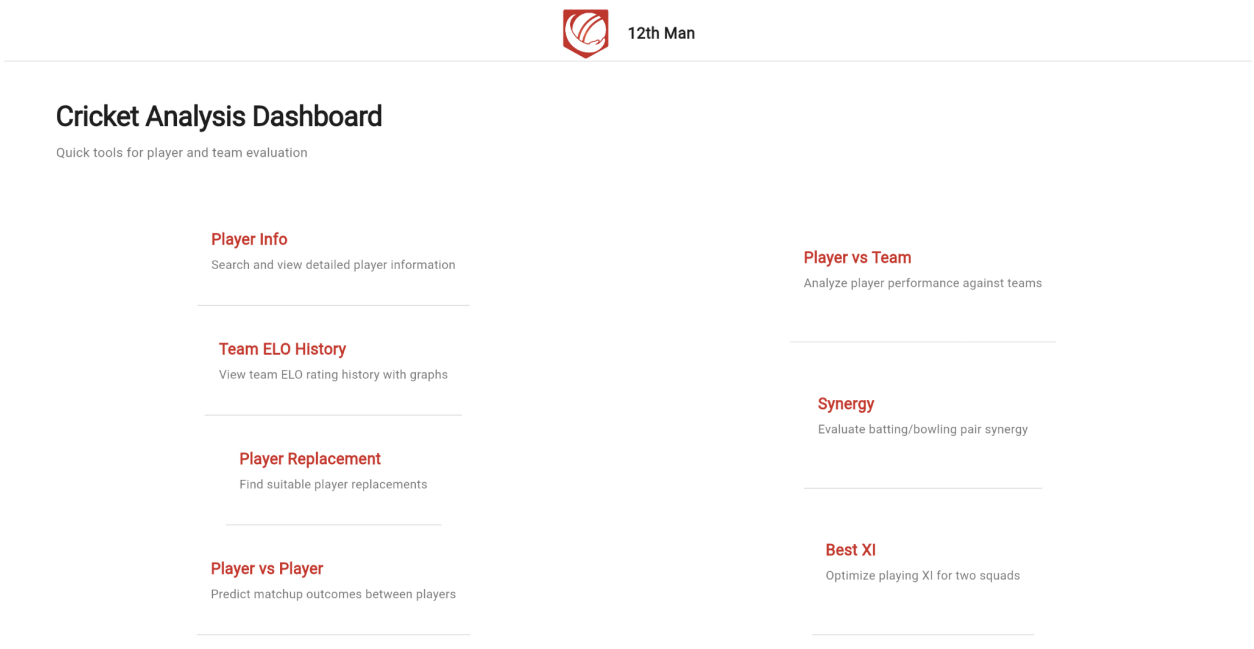


Fig. 6.4. Landing Page of the frontend

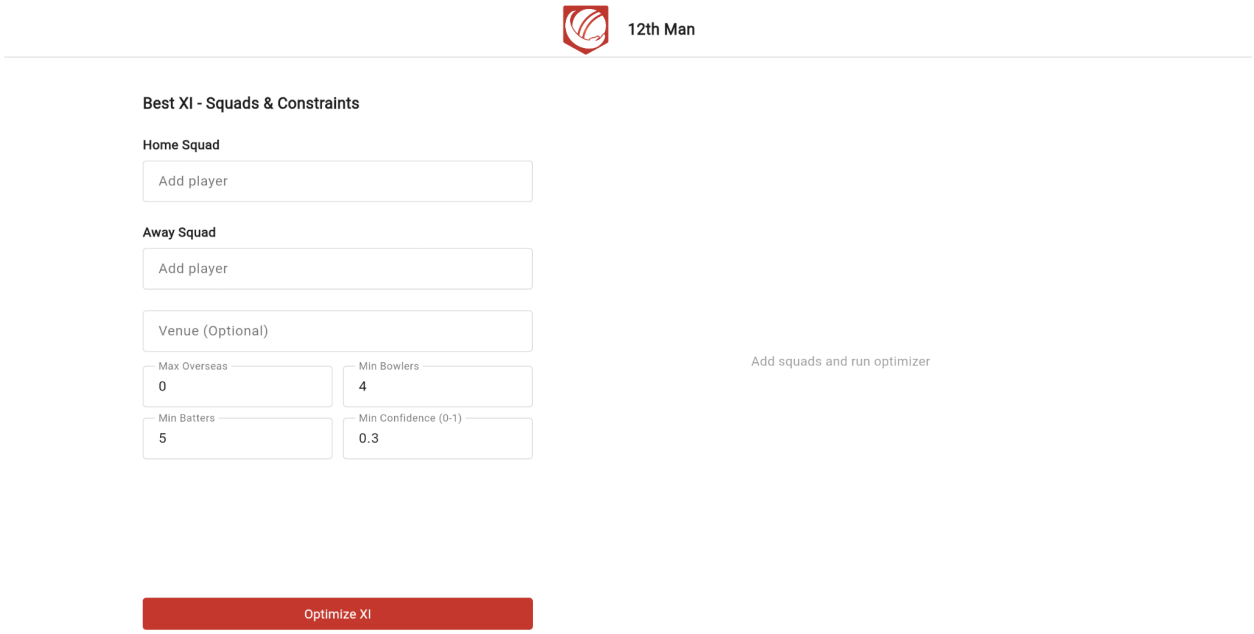


Fig. 6.5. Best XI Optimizer Page of the frontend

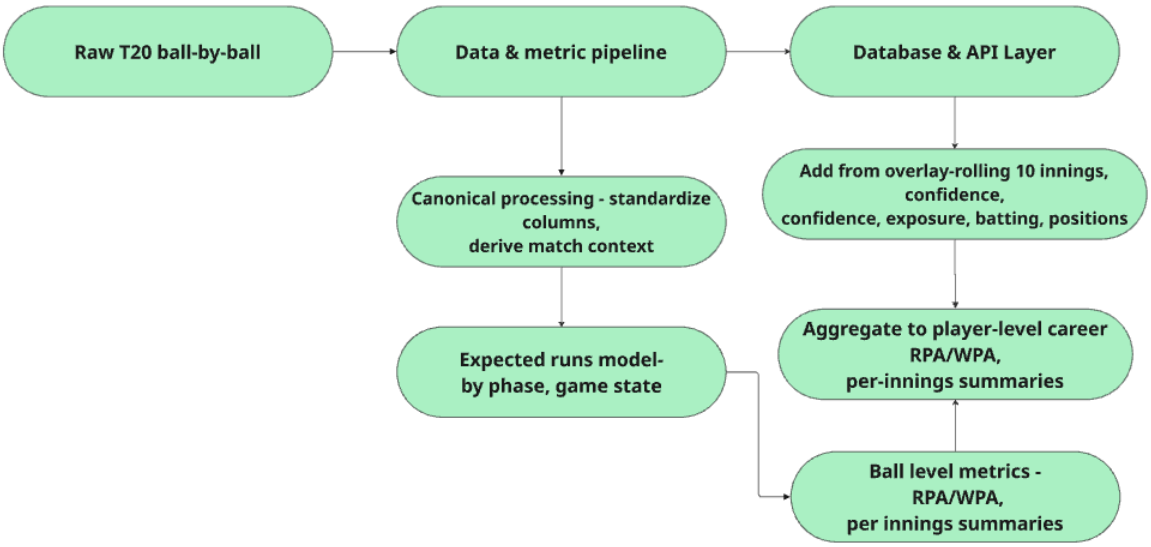


Fig. 6.6. Working of the application model(Backend working).

6.2 Implementation Formulae

Table I Formulae Table

Metric/Concept	Formula/Calculation	Explanation
1) Win Probability Added (WPA)	$WPA_i = WP_{after,i} - W_{before,i}$	Change in win probability for each ball i
2) Runs Per Average / Runs above Expected (RPA)	$E[runs_i \mid C_i] = \mu(C_i)$ $RPA_{raw,i} = runs\ of\ bat_i - E[runs_i \mid C_i]$	"Runs above expected" per delivery, benchmarked against a context-specific expected runs model.
3) Aggregating to player level (Z-scores)	$RPA_{bat} = \frac{1}{N_{bat}} \sum_{i=1}^{N_{bat}} RPA_{z,i}^{bat}$ $RPA_{bowl} = \frac{1}{N_{bowl}} \sum_{i=1}^{N_{bowl}} RPA_{z,i}^{bowl}$ $WPA_{bat} = \frac{1}{N_{bat}} \sum_{i=1}^{N_{bat}} WPA_{i,bat}^{*}$ $WPA_{bowl} = \frac{1}{N_{bowl}} \sum_{i=1}^{N_{bowl}} WPA_{i,bowl}^{*}$ *bat_rpa_z, bowl_rpa_z, bat_wpa_z, bowl_wpa_z*	Ball-level RPA/WPA are averaged and standardized (z-scored) across all players for common comparison.
4) Player value and replacement (4-D vector)	$v = [bat_rpa_z,\ bowl_rpa_z,\ bat_wpa_z,\ bowl_wpa_z]$	A vector representing a player's impact profile across the four core dimensions.
4) Cosine similarity for replacement	$\cos(\theta) = \frac{v_A \cdot v_B}{\|\|v_A\|\| \, \|\|v_B\|\|}$	Returns a similarity between -1 and 1; higher = more similar overall impact profile, used to find like-for-like replacements.

5) Career value (Batter example)	$CareerValue = 0.40 \cdot bat_{rpa_z} + 0.4 \times bat_{wpa_z} + 0.1 \times bowl_{rpa_z} + 0.10 \times bowl_{wpa_z}$	Collapses the 4-D vector into one role-aware career rating (e.g., higher weight on batting for a batter). All-rounder uses 0.25 on each.
5) Career value (Bowler example)	*CareerValue=0.10 · bat_rpa_z+0.10 · bat_wpa_z+0.40 · bowl_rpa_z+0.40 · bowl_wpa_z*	Collapses the 4-D vector into one role-aware career rating (e.g., higher weight on bowling for a bowler).
6) Combined Value (Long-term + Form)	$Value_{comb} = 0.8 \cdot CareerValue + 0.2 \cdot FormZ$	A composite value combining 80% long-term quality CareerValue and 20% recent performance (FormZ - z-score of last K innings).
7) Confidence (data volume)	$FinalValue = Value_{comb} \cdot Confidence$	Confidence calculation involves $balls_{faced}$+ $balls_{bowled}$ E relative to a threshold (T). Small-sample players are down-weighted by a confidence score closer to 0.
8) PVP (Player vs Player) - Matchup advantage	Uses historical ball-by-ball stats between the pair (Expected Runs Per Ball, Dismissal Rate), adjusted by their career baselines.	Quantifies the edge one player has over another in a direct matchup based on historical data.
9) Batting Synergy	$Synergy_{bat} = RPA_{together} - \frac{RPA_{1,solo} + RPA_{2,solo}}{2}$ (A simplified interpretation of the concept)	Positive score means two batters outperform their individual baselines when batting together; negative

		means they underperform.
9) Bowling Synergy	Analogous to batting synergy, using economy/RPA.	Positive if their combined economy/RPA improves when they operate in the same match/phase.

7.EXPERIMENTATION & RESULTS

7.1 Datasets

The dataset that has been derived for the use of the models has been obtained from multiple sources. The major source of ball-by-ball match data has been Cricksheet [8], which provided detailed and structured data. This raw data was then processed to formulate and obtain WPA, RPA and Team ELO. Another source of data was “cricket-player-worldwide-target” [9] from Kaggle which helped in determination of player’s nationality and batting-bowling style.

The following datasets were derived and used to train the models:

Table II WPA-RPA

Field name	**Description**
Match Id	Unique identification of a match
Innings	Inning of the said match
Over	Over of the innings
Ball	Ball of the running over
Batting Team	Name of the batting team
Bowling Team	Name of the bowling teams
Batter	Name of the batsman
Bowler	Name of the bowler
Runs off bat	Runs scored on that ball
Extras	Extras runs conceded
Is wicket	Ball conceded a bowled player
Win probability	Probability of win outcome
WPA	Win Probability Analysis Score
Date	Date of the match
League	Name of the league
Venue	Venue of the match being held

The WPA-RPA dataset records contain ball-by-ball data, including batting and bowling team data,players involved, runs scored, wickets and match context. It has calculated the field of WPA, quantifying batting and bowling events based on match outcome.

Table III Player form overlay

Field name	**Description**
Player	Name of the player
Form_bat_wpa	WPA of batting form
Form_bowl_wpa	WPA of bowling form
Form_total_wpa	Overall WPA combined
form_bat_z	Total WPA of last n matches- form_wpa
form_bowl_z	Total WPA of last n matches-forma_wpa
form_total	Total score in a specific form
form_confidence	Richness of a player in that specific form

Player form overlay stores the short term player analytics using WPA based metrics. It has rolling data of last n matches to take the current form of batters and bowlers into consideration and has a normalized z score for reliability.

Table IV Team ELO

Field Name	**Description**
Match Id	Unique Id of a match
Date	Date of the match
League	League/Form of the match
Team	Name of the team
Opponent	Name of the opponent team
Elo_after	Team's Elo rating after the match, representing updated team strength.
Elo-delta	Change in Elo rating resulting from the match outcome.
Confidence	Measure of reliability of the team's performance estimate based on match history.
Effective_elo	Adjusted Elo rating incorporating confidence and contextual match factors.
Matches_playes	No. of matches played
Result	Result/Outcome of the match

The team ELO dataset uses an elo rating framework to track the strength of the team. It records match wise updates, confidence levels and adjusts elo according to opponent elo and match outcome.

Table V Base Player Vector

Field Name	**Description**
Player	Name of the player
Role	Role in the match
Exposure	Measure of the player's involvement or participation level in matches.
Balls_faced	Total balls faced
Balls_bowled	Total bowls bowled/delivered
Career_bat_rpa_z	Total RPA gained in the batting
Career_bowl_rpa_z	Total RPA gained in the bowling
Career_bat_wpa_z	Total WPA in batting
Career_bowl_wpa_z	Total WPA in bowling
Inferred nationality	Nationality of the player
Batting_style	Style/Form of batting
Bowling_style	Style/Form of bowling

Base Player Vector dataset stores long term player attributes and career level performance indicators derived from WPA and RPA metrics. The dataset includes role classification, player nationality, participation of the player.

Table VI Total Player Value

Field Name	**Description**
Player	Name of the player
bat_rpa_raw	Raw RPA of batting form
bowl_rpa_raw	Raw RPA of bowling form
bowl_balls	Total balls bowled
bat_wpa_raw	Raw WPA for batting form
bowl_wpa_raw	Raw WPA for bowling form
ball_faced_wpa	WPA of balls faced by player

ball_bowled	No of bowls delivered
bat_rpa_eligible	Eligible for RPA w.r.t. batting
bowl_rpa_eligible	Eligible for RPA w.r.t. bowling
bat_wpa_eligible	Eligible for WPA w.r.t. batting
bowl_wpa_eligible	Eligible for WPA w.r.t. bowling
bat_rpa_z	Bat RPA score
bowl_rpa_z	Bowl RPA score
bat_wpa_z	Bat WPA Score
bowl_wpa_z	Bowl WPA Score
total_player_value	Total value of a player
last_match_played	Latest match played
teams_represented	Name of the team represented

The Total Player Value dataset aggregates the players batting and bowling WPA z-normalized scores to compute a unified score. It incorporates eligibility filters, recent participation, and historical performance to produce a final comparative player rating.

7.2 Testing

The testing results evaluate the performance of the Expected Runs (RPA) and Win Probability (WPA) models using standard regression and classification metrics.

Table VII Test Metrics

Model	Metric	Train	Val	Test	Notes
RPA	RMSE (runs)	17.21	-	25.12	Run prediction error; baseline for RPA.
	MAE (runs)	12.62	-	18.50	Average absolute prediction error.
	R^2	0.883	-	0.788	Variance explained by expected-runs model.
WPA	Log loss	0.4356	0.4428	0.4596	Cross-entropy of predicted win probabilities.
	Brier score	0.1408	0.1442	0.1476	Probability calibration error.
	ROC-AUC	-	-	0.8679	Discrimination between wins and losses.
	Accuracy	-	-	0.7883	Hard-threshold accuracy at 0.5 win probability.

The RPA model shows stable run prediction performance, with a test RMSE of 25.12 runs, MAE of 18.50 runs, and an R^2 value of 0.788, indicating that a significant portion of scoring variance is explained despite the inherent uncertainty of T20 matches.

The WPA model shows strong probabilistic performance, with a test log loss of 0.4596 and Brier score of 0.1476, reflecting good calibration. A ROC-AUC of 0.8679 and accuracy of 0.7883 confirm effective discrimination between winning and losing match states.

Overall, both models generalize well to unseen data and provide reliable outputs for downstream analytics such as PvP, PvT, Best XI selection, and match outcome prediction.

Comparison with Existing Work:

The results align with established sports analytics benchmarks, where expected runs models typically achieve R^2 values between 0.70 and 0.85, calibrated win probability models report log loss between 0.40 and 0.50 with Brier scores around 0.14-0.18, and ROC-AUC values above 0.85 are considered strong for match outcome prediction.

Conclusion for Testing:

The testing results demonstrate that the proposed RPA and WPA models perform within industry accepted analytical standards and are comparable to established sports analytics systems. The balance between accuracy, calibration, and interpretability makes the models suitable for real-world cricket analysis applications.

7.3 Results

Case 1 : Synergy Analysis

Analysis from a reputed website

Analysis from the system

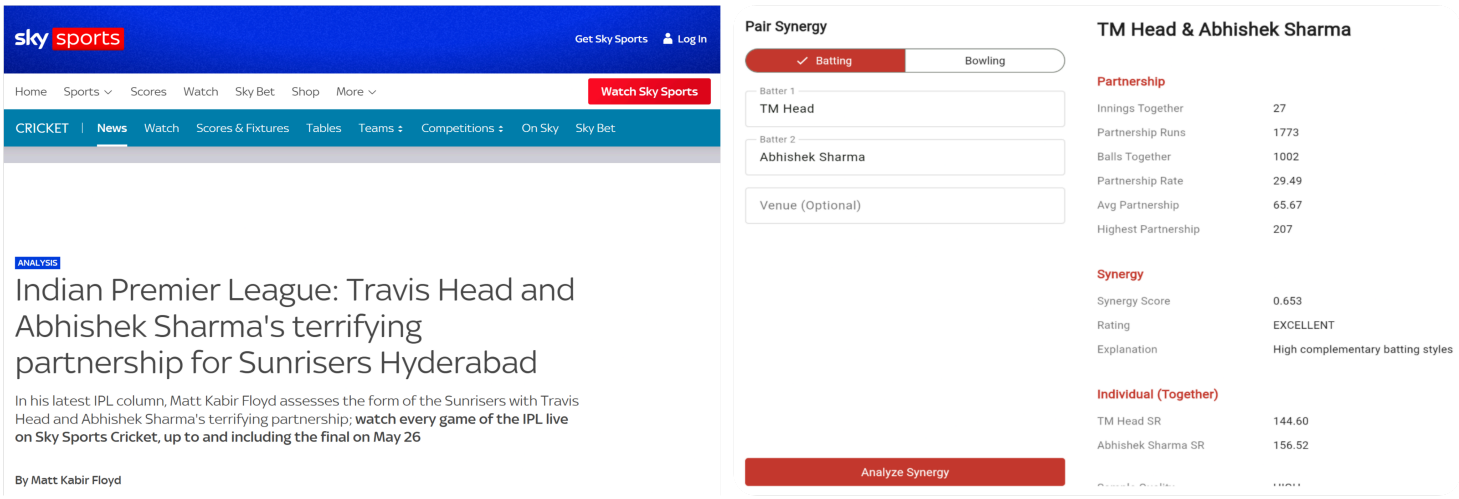


Fig. 7.1. Comparing the Synergy Analysis of the model with an article from “SkySports”[10].

Case 2 : Best XI

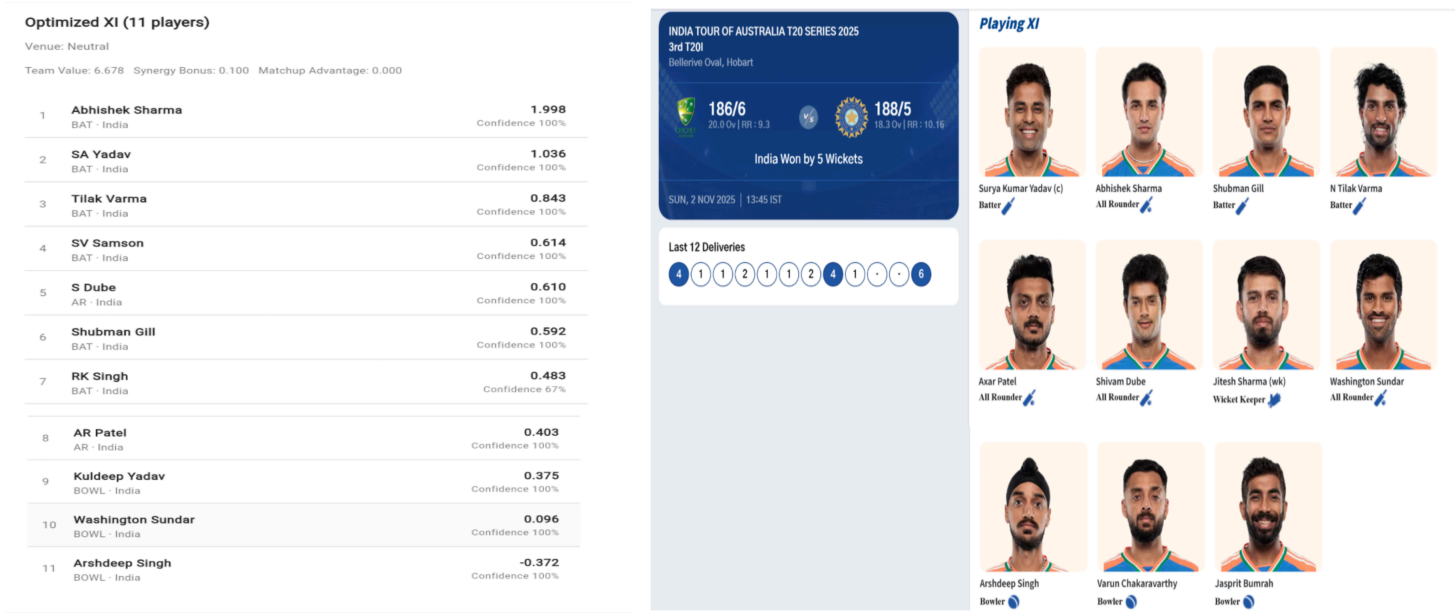


Fig. 7.2. Comparing the Best XI provided by the model with the actual playing XI.

Case 3 : Best Replacement

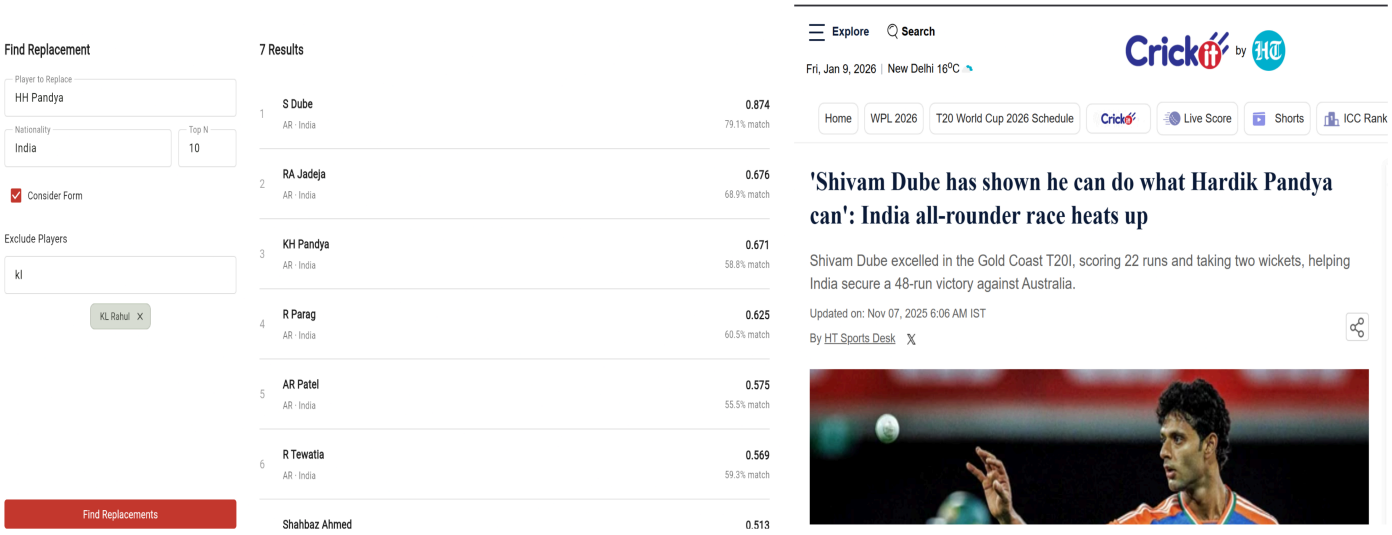


Fig 7.3. Comparing the systems best replacement with an article from “Hindustan Times”[11].

In fig 7.1 , the system correctly analyses the batting duo of Abhishek Sharma and TM Head as an excellent partnerships which rightly parallels the real world notion of them being one of the most explosive pairs playing together[10].

In fig 7.2 , the system put together a match X1 with the inputs of the same squad selected of the series and available for that particular match along the opposition squad with dated 2nd November , 2025 . The system manages to get 8/11 players correctly in that XI .

In fig 7.3 , in a scenario where HH Pandya gets injured while playing for the Indian team , the team has to find a replacement with similar skillsets . The system accurately lists S Dube as an ideal candidate which can be further confirmed from the article[11] . Both the cricketers and similar explosive batting styles and part - time fast bowling capabilities.

8. CONCLUSION

This project shows that it is possible to summarise a player's T20 impact in a single, interpretable value space that still respects role, context and uncertainty, instead of relying on disconnected statistics like average or strike rate.

The framework also demonstrates how short-term form and data-volume-aware confidence can be layered on top of long-term value so that decision-making naturally balances "track record" against "current touch" while avoiding small-sample noise.

Because all downstream use cases (replacement, matchup, synergy, XI selection) read from the same underlying value representation, decisions at different levels of detail remain consistent with each other; this is a conceptual shift from ad-hoc, module-by-module heuristics to a unified decision language for T20 cricket.

Finally, the work illustrates a general blueprint for sports analytics systems: start from granular event data, build calibrated context-aware metrics, embed entities in a low-dimensional but interpretable space, and expose that space through a lightweight data store and services layer so that coaches, analysts and tools can all consume the same coherent view of value.

9. REFERENCES

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APPENDIX A

Turnitin Similarity Report

The project report was evaluated for originality using the Turnitin plagiarism detection system. The similarity index is shown below.

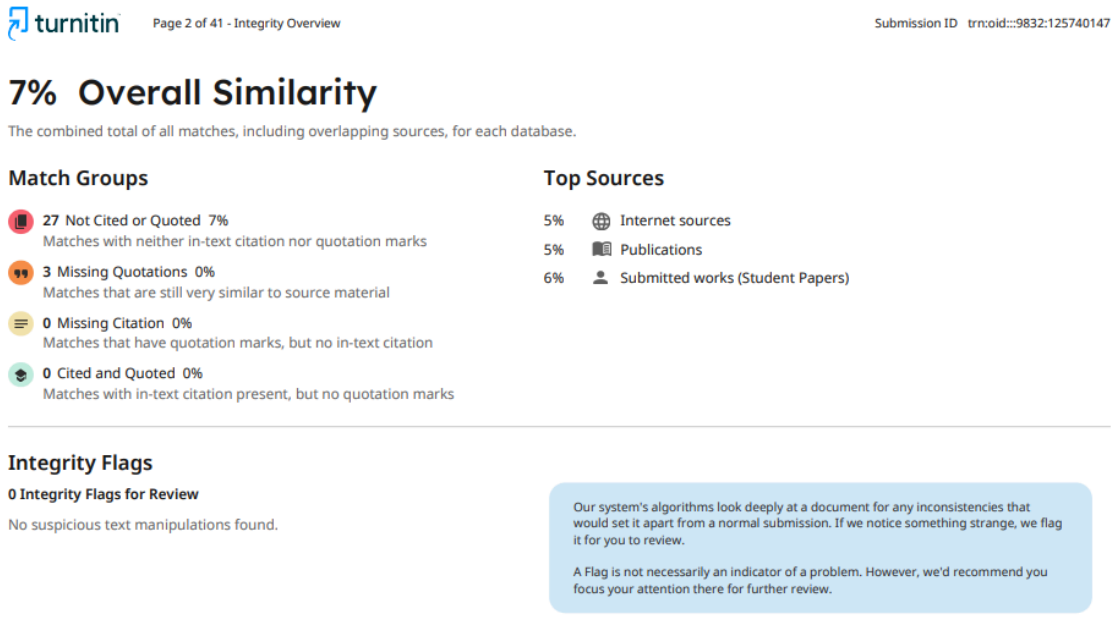


Fig. A.1. Turnitin Similarity Report Summary

Turnitin AI Writing Report

The project report was additionally evaluated using Turnitin's AI writing detection tool to assess the likelihood of AI-generated content. The AI writing report provides an estimated percentage based on linguistic and statistical patterns. The results is shown below

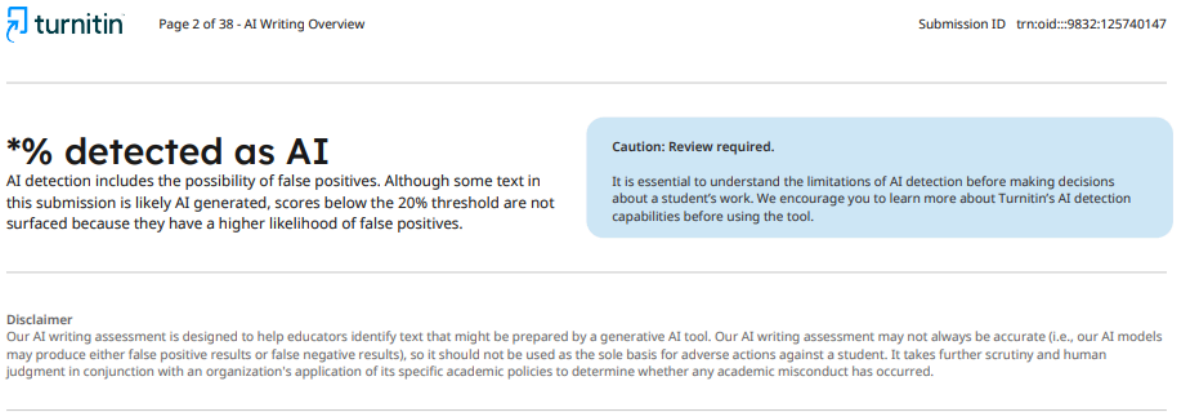


Fig. A.2. Turnitin AI Writing Report Summary